

# GroundingDINO — Findings Report v2

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## Overview

This report summarizes exploration of Grounding DINO, an open-vocabulary object detection model, as a potential first-stage component in a deepfake-detection pipeline — specifically, for locating faces or facial regions before a dedicated forensic manipulation-detection model is applied downstream. This is an updated version of the initial findings report, incorporating three additional test conditions requested in supervisor feedback: an expanded occluded-face category, a new small/low-resolution-face category, and an initial (non-forensic) deepfake-image sample.

## Model Summary

Grounding DINO addresses a core limitation of conventional object detectors: fixed category lists set during training. Instead of a closed vocabulary, it accepts a natural-language text prompt alongside an image, and grounds objects in the image to phrases in that prompt, allowing detection of arbitrary categories at inference time without retraining.

The model takes two inputs (an image and a text prompt) and fuses them throughout the detection pipeline rather than only at a final classification step, which lets language guide which image regions the model attends to. Its output is a set of bounding boxes, each paired with a confidence score and a phrase drawn from the input prompt describing what was detected.

Two thresholds control the output: the box threshold sets the minimum confidence a candidate box must reach to be kept at all, while the text threshold determines which words in the prompt are matched closely enough to a region to appear in its label. Together they let a user tune the tradeoff between missing real objects and surfacing noisy or duplicate detections.

## Setup

### Environment

| Component    | Version / Detail           |
|--------------|----------------------------|
| Python       | 3.10                       |
| PyTorch      | 2.13.0                     |
| Transformers | 4.37                       |
| CUDA         | N/A (macOS, no NVIDIA GPU) |
| GPU / Chip   | Apple M2                   |
| OS           | macOS 26.6                 |

Inference was run in `--cpu-only` mode rather than via MPS (Metal), since Grounding DINO's MPS path has known compatibility issues (e.g. `torch.roll` is unsupported on MPS without `PYTORCH_ENABLE_MPS_FALLBACK=1`, and `cumsum` lacks `int64` support, affecting accuracy), and CPU

mode benchmarks comparably fast in community tests. The official Swin-T checkpoint was downloaded and basic inference was validated via the repository's official demo script before building a custom batch pipeline.

## Methodology

A custom batch-inference script runs Grounding DINO across a folder of images with configurable prompt and threshold values via CLI arguments, saving annotated images and a predictions.json file containing per-image bounding box coordinates, matched phrases, confidence scores, and inference time. Across the original runs, inference averaged 3.37 seconds per image in CPU-only mode on the Apple M2.

## Test Image Set (v2)

Following feedback that the occluded category was too thin (2 images) to draw confident conclusions from, and that a small/low-resolution face condition was missing entirely, the test set was expanded. The multiple-faces category also gained one new image this round (see Section 5.2).

| Category                      | Count     | Description                                                                            |
|-------------------------------|-----------|----------------------------------------------------------------------------------------|
| Single frontal face           | 10        | LFW via Hugging Face mirror logasja/lfw                                                |
| Multiple faces                | 5 (was 4) | +1 new: one additional dense crowd scene (Unsplash)                                    |
| Profile / angled face         | 5         | Stock photos (Unsplash / Pexels)                                                       |
| Occluded / blurred face       | 7 (was 2) | +5 new: masks, hands, hair, sunglasses+scarf, partial crops (Unsplash/Pexels)          |
| Small / low-resolution face   | 6 (new)   | Thumbnail-resolution downscales (32–90px long edge) of already-curated photos          |
| Deepfake sample (placeholder) | 12 (new)  | Kaggle manjilkarki/deepfake-and-real-images, Test/Fake split — non-forensic substitute |

small\_lowres images are thumbnail-resolution downscaled copies of already-curated photos from single/profile/occluded (long edge 32–90px), isolating resolution as the sole variable rather than introducing new subjects/poses at the same time. The deepfake sample is an openly-downloadable, non-forensic Kaggle set used as a placeholder pending FaceForensics++/RWDF-23/FakeAVCeleb access (see Dataset Access Status below).

# Results

## 5.1 Prompt Experiments (Original Categories, Recap)

- “face . eyes . mouth” consistently produced more boxes than “human face” across every category (e.g. single: 3–11 boxes/image vs. 1–4 boxes/image) — expected, since the .-separated prompt creates three distinct grounding targets (face, eyes, mouth) rather than one box per face.
- “real face” and “manipulated face” tracked closely with “human face” in nearly all cases, indicating the model is not meaningfully responding to the “real”/“manipulated” qualifier — it continues to primarily match on “face.”
- The profile and (original 2-image) occluded categories were largely stable at 1 box/image across face-related prompts.
- “person” produced comparable box counts to “human face” on profile/occluded samples, but diverged on multiple-faces and single-face categories — suggesting “person” and “human face” generalize differently to crowds vs. individuals.

## 5.2 Multiple Category (Expanded, 4 → 5 images)

One new image was added to the multiple-faces category — a large, genuinely crowded stadium/rally photo (hundreds of people, individual faces roughly 30–40px) — bringing the category from 4 to 5 images. At default thresholds (box=0.35/text=0.25), this new image returned zero detections for four of the five standard prompts:

| Prompt              | Boxes on new crowd photo      |
|---------------------|-------------------------------|
| human face          | 0                             |
| person              | 0                             |
| face . eyes . mouth | 0                             |
| real face           | 0                             |
| manipulated face    | 1 (score 0.44 — likely noise) |

For comparison, the original multiple-faces outlier from the first round (nicholas-green, a smaller group) still returns 18 boxes on “human face” at the same threshold (see Section 5.6). This is a different failure mode than the duplicate-box behavior documented for that image — rather than over-detecting a crowded scene, the model misses it entirely once faces get small/dense enough. As the threshold sweep in Section 5.7 shows, this isn't recoverable by threshold tuning either.



Figure D. New multiple-faces image at the most permissive threshold tested (box=0.25/text=0.20) — 2 low-confidence boxes (0.296, 0.264) are visible if you look closely; at default and higher thresholds this image returns 0 boxes despite hundreds of visible faces.

### 5.3 Occluded Category (Expanded, 2 → 7 images)

With six new images added — covering masks, hands covering the face, hair covering the face, and sunglasses+scarf combinations — detection at default threshold (box=0.35/text=0.25, prompt “human face”) was highly consistent: 6 of 7 images produced exactly 1 box. The one exception was a hair-covering-face image, which produced 2 overlapping boxes at different confidences (0.60 and 0.39) — suggesting the model can produce a duplicate detection when hair partially bisects, rather than fully occludes, the visible face region.



Figure A. Occluded category example — hair partially covering the face, prompt “human face”, default thresholds. This image produced 2 overlapping boxes (0.60, 0.39) rather than the single box typical of the rest of the expanded category.

### 5.4 Small / Low-Resolution Category (New, 6 images)

Downscaling existing single-subject photos (single, profile, occluded) to thumbnail resolution (32–56px long edge) did not break detection: all three produced exactly 1 box, at confidence scores (0.57–0.82) comparable to their full-resolution originals. Downscaled multi-subject group photos (90px long edge) still detected multiple faces (4–5 boxes at default threshold) but at reduced count and confidence relative to the full-resolution source



— partial rather than complete degradation under combined multi-subject + low-resolution conditions.



Figure B. small\_lowres example, shown enlarged from its native 90×60px — a downsampled group photo, prompt “human face”, default thresholds. 5 of the original group’s faces were still detected at this resolution.

## 5.5 Deepfake-Sample Prompt Results (Exploratory)

The five standard prompts were run against the 12-image Kaggle placeholder sample (Test/Fake split, manjilkarki/deepfake-and-real-images) at default thresholds:

| Prompt              | Avg boxes/img | Total boxes (12 img) | Notes                                                                              |
|---------------------|---------------|----------------------|------------------------------------------------------------------------------------|
| human face          | 1.33          | 16                   | baseline                                                                           |
| person              | 1.58          | 19                   | slightly more than “human face”                                                    |
| face . eyes . mouth | 4.58          | 55                   | expected — separate targets, matches original pattern                              |
| real face           | 1.25          | 15                   | tracks closely with “human face”                                                   |
| manipulated face    | 1.17          | 14                   | tracks closely with “human face”; 1 zero-detection (only one across all 5 prompts) |

“Real face” (1.25) and “manipulated face” (1.17) both sit in the same range as plain “human face” (1.33) — not meaningfully lower or differently distributed. This replicates the original-round finding on ordinary photos, now on a deepfake-labeled (if non-forensic) sample: the model is not detecting anything forensically different between the qualifiers.



Figure C. Deepfake-sample example (Kaggle Test/Fake), prompt “human face”, default thresholds, score 0.83.

**Caveat:** this is an openly-downloadable Kaggle image set, not a forensic deepfake-detection benchmark (FaceForensics++/RWDF-23/FakeAVCeleb). Results here are exploratory only — not a reliable signal that Grounding DINO, or the “real”/“manipulated” prompt wording, can distinguish genuine from manipulated media.

## 5.6 Threshold Sensitivity — Original Categories (Recap)

Three representative images (one each from multiple, single, and profile, selected for high or borderline box counts) were re-run at three box/text threshold combinations:

| Image (category)                           | 0.25/0.20 | 0.35/0.25          | 0.45/0.30 | Interpretation                                                                               |
|--------------------------------------------|-----------|--------------------|-----------|----------------------------------------------------------------------------------------------|
| lfw_02 (single)                            | 4 boxes   | 4 boxes, identical | 1 box     | Extra boxes at low threshold are genuine background faces, not duplicates.                   |
| nicholas-green (multiple)                  | 26 boxes  | 18 boxes           | 1 box     | Heavy duplication at low threshold; highest threshold undercounts a genuinely crowded scene. |
| pexels-moh-adbelghaffar (profile/occluded) | 3 boxes   | 1 box              | 0 boxes   | Highest threshold misses the face entirely.                                                  |

## 5.7 Threshold Sensitivity — Multiple, Occluded, and Small/Low-Resolution

The same three threshold combinations were run against the expanded/new categories:

| Image (category)                            | 0.25/0.20              | 0.35/0.25          | 0.45/0.30 | Interpretation                                                                                                                                                                                                           |
|---------------------------------------------|------------------------|--------------------|-----------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| neelakshi-singh (multiple, new dense crowd) | 2 boxes [0.296, 0.264] | 0 boxes            | 0 boxes   | Even the most permissive threshold recovers only 2 very-low-confidence boxes out of hundreds of visible faces. Not a threshold-tuning problem — the scene is effectively invisible to the model at every setting tested. |
| yogesh-rahamatkar (occluded)                | 2 boxes [0.386, 0.399] | 2 boxes, identical | 0 boxes   | Only occluded image sensitive to threshold — both detections score below 0.45. All other occluded images stayed at 1 box across all three thresholds.                                                                    |
| thumb_multiple_02 (small_lowres)            | 8 boxes                | 5 boxes            | 1 box     | Sharpest decline of any small_lowres image — a downscaled group photo loses most detections well before the highest threshold.                                                                                           |
| thumb_single_lfw03 (small_lowres)           | 2 boxes                | 1 box              | 1 box     | Extra box at lowest threshold is a low-confidence duplicate that disappears by 0.35/0.25; otherwise stable.                                                                                                              |

Two distinct patterns emerge. The expanded occluded category and the single/profile-derived small\_lowres images behave like the original categories: threshold mainly trades off duplicate/spurious boxes against missed genuine ones, and 0.35/0.25 remains a reasonable general-purpose default. But the dense-crowd case doesn't respond to threshold tuning the same way — lowering the threshold recovers duplicate boxes on a scene the model already detects (as with the original nicholas-green outlier, Section 5.6), but does almost nothing for a scene it fundamentally failed to detect in the first place. Threshold sweeping is not a fix for the crowd-density failure mode documented in Section 5.2.

## Dataset Access Status (deepfake condition)

| Dataset                | Access                          | Status                                         |
|------------------------|---------------------------------|------------------------------------------------|
| FaceForensics++        | Request form                    | Pending (originally requested; follow-up sent) |
| RWDF-23 (DASH Lab)     | Google Form                     | Requested                                      |
| FakeAVCeleb (DASH Lab) | Google Form + license agreement | Requested                                      |

None of the three are approved as of this report. The Kaggle sample used in Section 5.5 is a stopgap, not a replacement, for genuine forensic-quality deepfake frames once one of these is granted.

## Limitations

- **Detection can fail completely, not just duplicate/over-detect, once a scene is dense/small enough**, and this failure does not resolve with threshold tuning (Sections 5.2, 5.7). This is a more significant limitation than the duplicate-box issue alone for any pipeline relying on this as a first-stage face locator.
- **Deepfake content still non-forensic**. The “real face”/“manipulated face” results reflect an openly-downloadable Kaggle set, not FaceForensics++/RWDF-23/FakeAVCeleb; results should be re-evaluated once genuine forensic-quality frames are available.
- **Occluded (7) and multiple (5) category sizes are improved but still modest** — findings remain directional, not statistically robust.
- **No single threshold setting is universally correct** across categories or resolutions; a production pipeline would need either a per-scene-type threshold policy or an alternative to a single global cutoff, such as adaptive thresholding.
- **Duplicate box proposals** at lower thresholds continue to appear (now also observed on an occluded hair-covering example) — raw detection output requires post-processing (e.g. non-max suppression tuning) before use as input to a downstream forensic model.
- **Small/low-resolution degradation is partial, not binary**, for the conditions tested here — single-subject detection was robust down to ~32px, while multi-subject detection degraded gradually rather than failing outright.

## Challenges

- Function signature mismatches between the CLI demo script and the underlying groundingdino.util.inference API (load\_model and predict each have independent device parameters with different defaults), requiring direct inspection of the source rather than assuming consistent behavior.
- A transformers version incompatibility (AttributeError: 'BertModel' object has no attribute 'get\_head\_mask') required pinning to transformers==4.37.
- The threshold-sweep script initially failed silently: a bash idiom (set -- \$combo) used to split a “box text” pair does not word-split the same way in zsh (macOS's default shell) as in bash, so all sweep runs errored out with an invalid-float-value argparse error before writing any output. Fixed by switching to shell-portable parameter-expansion splitting (`${combo%%:*} / ${combo##*:}`).

## Next Steps

- Investigate the crowd-density detection failure further (Section 5.2) — e.g. whether tiling the image before inference, or running at a higher input resolution, recovers detections that a single global threshold cannot.
- Incorporate genuine deepfake dataset frames once FaceForensics++/RWDF-23/FakeAVCeleb access is granted, and re-run the “real face”/“manipulated face” prompt comparison on that content in place of the Kaggle placeholder.

- Investigate non-max suppression or other post-processing to address duplicate box proposals, now observed in both crowded and occluded (hair-covering) cases.
- Consider a per-category or adaptive threshold policy rather than a single global default, given how differently occluded, small/low-resolution, and multi-subject images respond to threshold changes.